[Supplemental Material] Portolan: A Tail-Calibrated Benchmark for City-Scale Cross-Camera Trajectory Systems

I. Complete Proof of Proposition 1

Proposition 1 (Cross-Camera Moment Calibration): For even $d \geq 4$, unit prototype $\mathbf{u}_v \in \mathbb{S}^{d-1}$, independent camera rotors $\mathbf{R}_c = \mathbf{W}_0 \operatorname{diag}(\mathbf{G}(\phi_{c,1}), \ldots, \mathbf{G}(\phi_{c,d/2}))\, \mathbf{W}_0^\top$ $(c_i \neq c_j)$ with 2×2 rotations $\mathbf{G}(\phi) \in SO(2)$, independent angles $\phi_{c,m} \sim \mathcal{N}(0, \sigma_\phi^2)$, $\mathbf{W}_0 \in O(d)$, and conditionally independent sightings $\mathbf{e}_i \sim \mathrm{vMF}(\mathbf{R}_{c_i}\mathbf{u}_v, \kappa_{c_i})$, $\mathbf{e}_j \sim \mathrm{vMF}(\mathbf{R}_{c_j}\mathbf{u}_v, \kappa_{c_j})$, the expected cross-camera cosine satisfies

$$\mathbb{E}[\mathbf{e}_i^\top \mathbf{e}_j] = A_d(\kappa_{c_i}) A_d(\kappa_{c_j}) e^{-\sigma_\phi^2}, \tag{1}$$

where $A_d(\kappa) = I_{d/2}(\kappa)/I_{d/2-1}(\kappa)$ is the Bessel ratio. Whenever $0 < \mu_{\mathrm{pos}} < e^{-\sigma_\phi^2}$, setting $\kappa_{c_i} = \kappa_{c_j} = \kappa_{\mathrm{Prop1}}$ with

$$\kappa_{\mathrm{Prop1}} = \frac{d-1}{2\big(1 - \sqrt{\mu_{\mathrm{pos}} \exp(\sigma_\phi^2)}\big)} \tag{2}$$

matches target cross-camera similarity μ_{pos} up to $O_d(\kappa_{\mathrm{Prop1}}^{-2})$ as $\mu_{\mathrm{pos}} \to e^{-\sigma_\phi^2}$.

Proof: Let $\boldsymbol{\mu}_i = \mathbf{R}_{c_i}\mathbf{u}_v \in \mathbb{S}^{d-1}$. Any unit vector $\mathbf{e}_i \sim \mathrm{vMF}(\boldsymbol{\mu}_i, \kappa_{c_i})$ admits the tangent-normal decomposition $\mathbf{e}_i = t_i \boldsymbol{\mu}_i + \sqrt{1 - t_i^2}\, \boldsymbol{\xi}_i$, where the radial projection $t_i = \mathbf{e}_i^\top \boldsymbol{\mu}_i \in [-1, 1]$ is independent of the tangential direction $\boldsymbol{\xi}_i$ uniformly distributed on the orthogonal equator $\mathbb{S}^{d-2} \perp \boldsymbol{\mu}_i$. By equatorial symmetry, $\mathbb{E}[\boldsymbol{\xi}_i] = \mathbf{0}$, and by definition of the Bessel ratio, $\mathbb{E}[t_i] = A_d(\kappa_{c_i})$, giving $\mathbb{E}[\mathbf{e}_i \mid \mathbf{R}_{c_i}, \mathbf{u}_v] = A_d(\kappa_{c_i})\mathbf{R}_{c_i}\mathbf{u}_v$.

Conditioning on $(\boldsymbol{\mu}_i, \boldsymbol{\mu}_j)$, conditional independence across sightings gives $\mathbb{E}[\mathbf{e}_i^\top \mathbf{e}_j \mid \boldsymbol{\mu}_i, \boldsymbol{\mu}_j] = A_d(\kappa_{c_i}) A_d(\kappa_{c_j}) \boldsymbol{\mu}_i^\top \boldsymbol{\mu}_j$. Writing $\tilde{\mathbf{u}}_v = \mathbf{W}_0^\top \mathbf{u}_v \in \mathbb{S}^{d-1}$ with 2-D blocks $\tilde{\mathbf{u}}_{v,m} \in \mathbb{R}^2$ and angle differences $\Delta\phi_m = \phi_{c_j,m} - \phi_{c_i,m} \sim \mathcal{N}(0, 2\sigma_\phi^2)$, orthogonality of $\mathbf{W}_0$ and cancellation of the skew-symmetric sine off-diagonals in each 2×2 rotation block $\mathbf{G}(\Delta\phi_m)$ yield

$$(\mathbf{R}_{c_i}\mathbf{u}_v)^\top (\mathbf{R}_{c_j}\mathbf{u}_v) = \sum_{m=1}^{d/2} \|\tilde{\mathbf{u}}_{v,m}\|_2^2 \cos \Delta\phi_m.$$

Because $\mathbb{E}[\cos \Delta\phi_m] = \exp(-\frac{1}{2}\operatorname{Var}(\Delta\phi_m)) = e^{-\sigma_\phi^2}$ and $\sum_{m=1}^{d/2} \|\tilde{\mathbf{u}}_{v,m}\|_2^2 = 1$, taking expectation over the plane angles gives $\mathbb{E}[\boldsymbol{\mu}_i^\top \boldsymbol{\mu}_j \mid \mathbf{u}_v] = e^{-\sigma_\phi^2}$. For common concentration $\kappa_{c_i} = \kappa_{c_j} = \kappa$, matching $\mathbb{E}[\mathbf{e}_i^\top \mathbf{e}_j] = \mu_{\mathrm{pos}}$ requires $A_d(\kappa) = \sqrt{\mu_{\mathrm{pos}} \exp(\sigma_\phi^2)}$. Substituting the high-concentration Bessel ratio expansion $A_d(\kappa) = 1 - \frac{d-1}{2\kappa} + O_d(\kappa^{-2})$ and solving for κ yields (2) up to $O_d(\kappa_{\mathrm{Prop1}}^{-2})$ for fixed d as $\mu_{\mathrm{pos}} \to e^{-\sigma_\phi^2}$. ■

II. Hardware Stage-Profiling Breakdown for `3D-Sim`

Table I summarizes the per-pass GPU latency breakdown used by the `3D-Sim` cost model in Sections V-A and V-B of the main paper on one NVIDIA RTX 4090 GPU.

TABLE I
HARDWARE MICROBENCHMARK BREAKDOWN FOR `3D-Sim` ON ONE NVIDIA RTX 4090 GPU (64 PASSES, 10 FRAMES/PASS).

Pipeline Stage	Latency / Pass	Share	Sample Time
1080p shader render + YOLOv8-m	21.51 ms	96.8%	1.377 s
ViT-B/16 TransReID extraction	0.71 ms	3.2%	0.046 s
Combined (45.00 **passes/s**)	22.22 **ms**	100.0%	1.422 **s**

Active GPU kernel buffer: 1,074.4 MB; extrapolated 5K-cam VRAM: 52,874.4 MB.

III. Out-of-Sample Calibration Seed Profiles

Table II reports the primary empirical moment and tail parameters stored in `data/real_seed_profiles.json`, measured strictly on the upstream corridor split of CityFlowV2 (`S01+S02`) and the held-out calibration trip split of VeRi-776.

TABLE II
EMPIRICALLY MEASURED OUT-OF-SAMPLE SEED PROFILES (`data/real_seed_profiles.json`).

Calibrated Moment / Tail Parameter	CityFlow-Seed	VeRi-Seed
Calibration split	9 cams (`S01+S02`)	20 cams (even trips)
Calibration passes	859 (240 vehs)	958 (137 trips)
Disjoint evaluation passes	2,178 (31 cams)	1,010 (136 trips)
Camera / split overlap	0% / 0%	20 shared / 0% trips
Cross-pass positive cosine (μ_{pos})	0.3837	0.4776
Positive cosine std. dev.	0.1955	0.2020
Random negative cosine	0.0141	0.0099
Hard-negative top-1 cosine tail	0.5133	0.4596
Natural hub sighting share	0.0093	0.0209
Near-view (P_{85}) vMF concentration	571.92	754.35
Camera rotation spread (σ_ϕ)	0.6734 rad	0.6072 rad
Congested log-transit std. dev.	0.4368	0.1576
Signal queue delay mean	18.47 s	22.78 s
Queue encounter share	0.3800	0.3784
Free-flow arterial speed	13.9 m/s	15.0 m/s